Reg. No. | | | | | | | | | | | [14CE65]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

DESIGN OF HYDRAULIC STRUCTURES

Time: 3 Hours **Maximum Marks: 70**

Instructions:

Answer any **FIVE** full Questions selecting atleast two from each part.
Use of Design charts is allowed
Assume any missing data suitably.

PART - A

1. a. Explain the hydraulic and structural requirement for Zone-1 (Top-Zone) and Zone-VI in a gravity dam. **04**

   b. Draw a neat diagram showing stress distribution around a circular hole in an infinite plate due to normal stress on the plate. **03**

   c. Define constant radius arch dam, variable radius arch dam and central angle arch dam in brief. Arrange them in ascending order with respect to their construction cost. **07**

2. a. Derive 2-D Laplace equation for seepage through earth dam. **04**

   b. List various methods of slices for stability analysis of finite slopes in earthern embankments. **03**

   c. The stability of downstream slope for an earth dam is to be investigated using the ordinary method of slices following data is given **07**

   i. $\gamma_{sat} = 20 kN/m^3$

   ii. $C' = 10 kN/m^2$

   iii. $\phi' = 27°$, $\alpha = 90°$

   iv. $r = 12m$

   The number of slices, its weight, uplift coefficient (r_4), slice angle (θ) are given in the following table

Slice No.	Weight in (w) KN	Spl. Ft. (R4) coefficient	Slice Angle (θ)
1	50	0.35	-15°
2	120	0.51	+4°
3	170	0.28	+21°
4	60	0.10	+50°

3. a. Comment on the Reinforcement provided near the crest region of a Ogee spillway. **05**

   b. Briefly explain the Hydraulic design procedure for a side channel spillway with a given discharge 'Q' and length (L) of the crest. **05**

   c. Describe the main function for providing on Intake structure and Trash rack at a river intake. **04**

4. a. List the parameters considered for obtaining the Economic size of a pipeline. **03**

   b. Mention various type of surge tank with a neat diagrams. **04**

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c. A conical draft tube of 5m length has a diameter of 2.0m at its top. Water discharges through it with flow rate of $20m^3/s$ and 1.2m/s velocity at the outlet. The pressure head at the top is 5.8 m of water (vacuum) and atmospheric pressure equals 10.3m of water. Neglecting friction loss, calculate the length of draft tube immersed in water and efficiency of draft tube. **07**

PART – B

5. a. What are the advantages of cylindrical gates over vertical lift gates. **04**

b. Write a short note on following. **10**

i. Butterfly valve

ii. Layout of a power house

iii. Needle valve

iv. Flow induced vibration and forces on a vertical lift gates

v. Howell – Bunger valve

6. a. What factors will you consider while selecting the alignment of the off taking channel. **04**

b. Design a cross-regulator for canal with the following data **10**

i. Discharge = 150 cumecs

ii. FSL u/s = 200.00 D/s = 199.8

iii. Bed level u/s = 197.50 D/s 197.30

iv. Bed width u/s = 55m D/S = 50m

v. Depth of water u/s = 2.50 D/s = 2.5

Assume a safe exit gradient of 1/6.

7. a. Fig. shows a hydraulic structure built on Fine sand (C=15). Determine **08**

i. Whether the percolation gradient is safe (Limit – 1/15).

ii. Uplift pressure and thickness at 'A', 'B' and 'C' at distances of 15m, 25m, 35m from the upstream end.

Use Bligh's and creep theory, G= 2.24

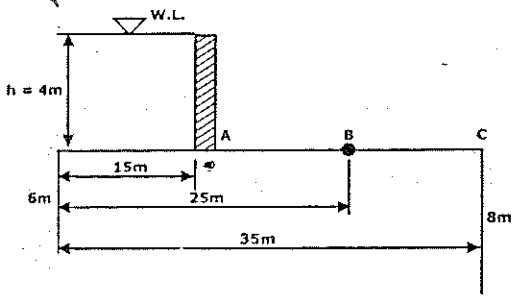

Fig. 7(a)

b. Explain Lane's weighted creep theory and compare it with Bligh's theory. **06**

8. a. Differentiate between Weir and Barrage **03**

b. Briefly explain design of a Weir with respect to surface flow, hydraulic jump and seepage flow characteristics. **08**

c. Draw a layout of a diversion head works showing its components parts. **03**